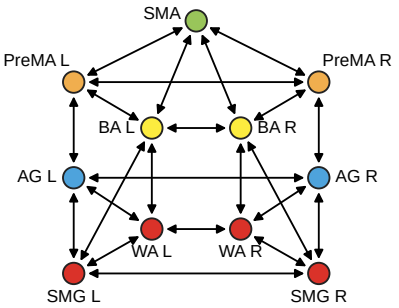
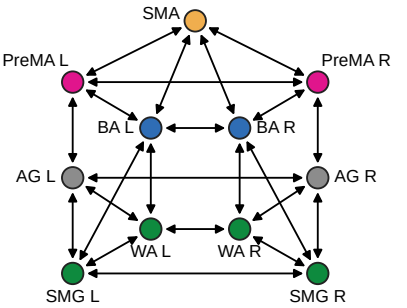
Symmetry analysis of dorsal stream baseline connectome

a)

Fibration symmetry = Group symmetry



5 Fibres



5 Orbits

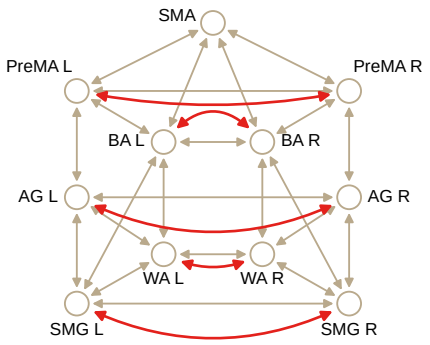
1: SMA
2: PreMA L
3: PreMA R
4: BA L
5: BA R
6: AG L
7: AG R
8: WA L
9: WA R
10: SMG L
11: SMG R

b)

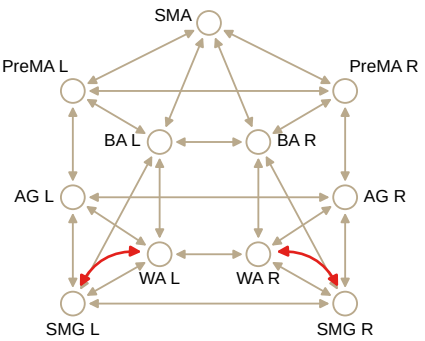
Automorphisms

$$\pi_1 = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 & 11 \\ \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \\ 1 & 3 & 2 & 5 & 4 & 7 & 6 & 9 & 8 & 11 & 10 \end{pmatrix} \quad \pi_2 = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 & 11 \\ \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \\ 1 & 2 & 3 & 4 & 5 & 6 & 7 & 10 & 11 & 8 & 9 \end{pmatrix} \quad \pi_3 = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 & 11 \\ \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \\ 1 & 3 & 2 & 5 & 4 & 7 & 6 & 11 & 10 & 9 & 8 \end{pmatrix}$$

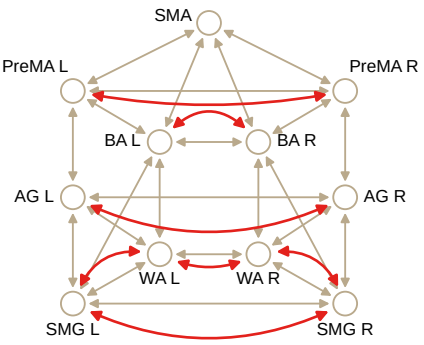
π_1



π_2



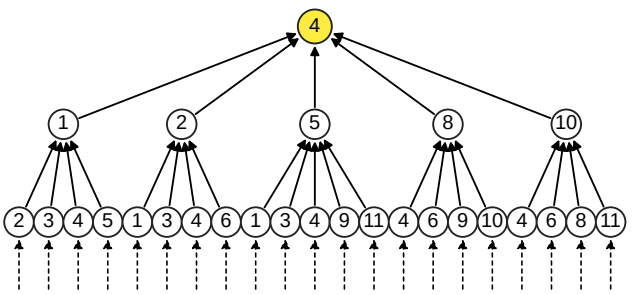
π_3



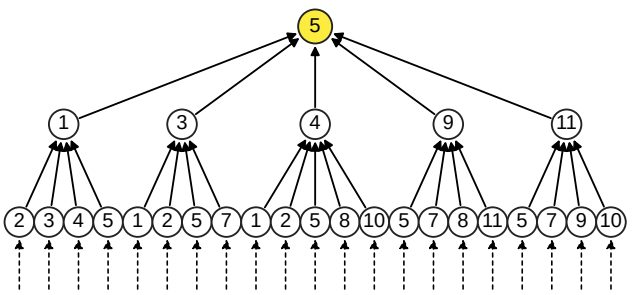
c)

Isomorphic Input Trees

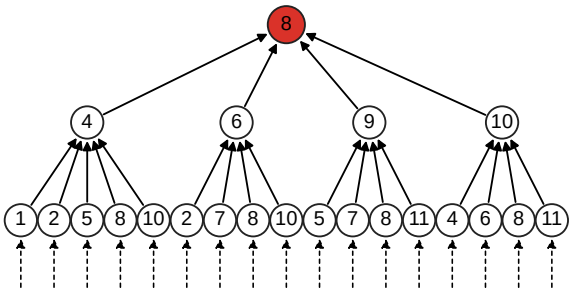
BA L



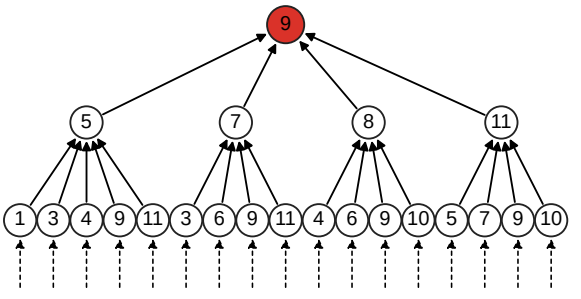
BA R



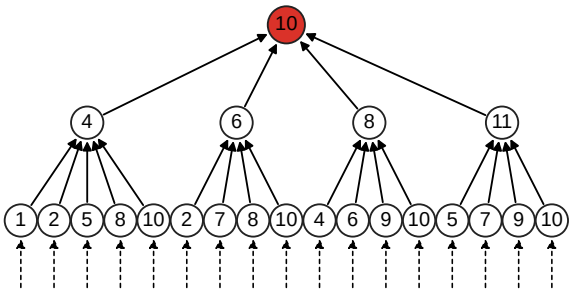
WA L



WA R



SMG L



SMG R

